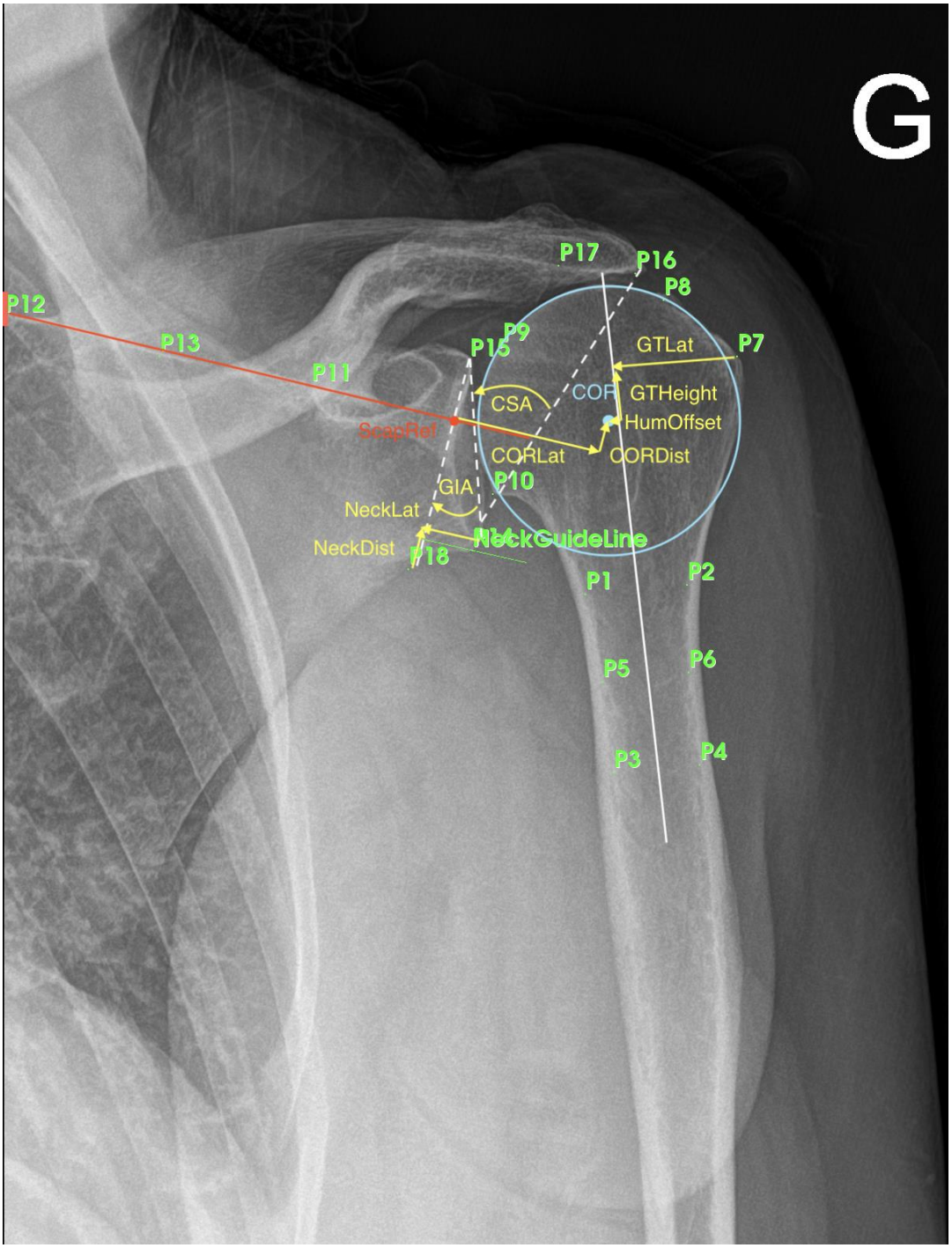
Point Placement Guide
Shoulder Analysis (Beta_angle_ana)

As a reminder, this extension enables the computation of shoulder morphological measurements on radiographs (or 2D images) based on precise anatomical landmarks. The accuracy of the results directly depends on the precision of point placement. Following, you will find a detailed description to help you place the fiducials nodes.



P1–P6 — Humeral diaphysis (proximal / mid / distal, medial / lateral) — inner cortex
Place six points along the *inner* cortical borders of the humeral shaft: proximal medial (P1) → proximal lateral (P2) → distal medial (P3) → distal lateral (P4) → mid-shaft medial (P5) → mid-shaft lateral (P6). P1-P3-P5 form the medial cortical border and P2-P4-P6 the lateral one. Spread the points as widely as possible along the shaft. The humeral axis is the centerline running midway between these two borders (angle bisector of the two least-squares cortical lines).

P7 — Latero-superior greater tuberosity (GT)
Place a point at the latero-superior edge of the greater tuberosity. It is used for GTHeight and GTLat (and for DistA / LatA in the reverse analysis).

P8, P9, P10 — Humeral head

- **P8: upper aspect of the humeral head, at its junction with the greater tuberosity.**
- **P9: lower head–calcar junction.**
- **P10: on the humeral head contour, roughly midway between P8 and P9.**

Together, P8, P9 and P10 define the circle of the humeral head, whose center is the center of rotation (COR). Space them as widely as possible along the contour for an accurate fit.

P11, P12, P13 — Fossa axis

- **P11: at the "bottleneck" where the coracoid process broadens away from the scapular fossa (lateral fossa – coracoid junction).**
- **P12: at the most medial point of the fossa visible on the image.**
- **P13: on the fossa cortex, roughly midway between P11 and P12. This third point lets the fossa axis be a least-squares fit rather than a simple two-point line.**

Together, P11, P12 and P13 define the fossa axis, used as the scapular reference line.

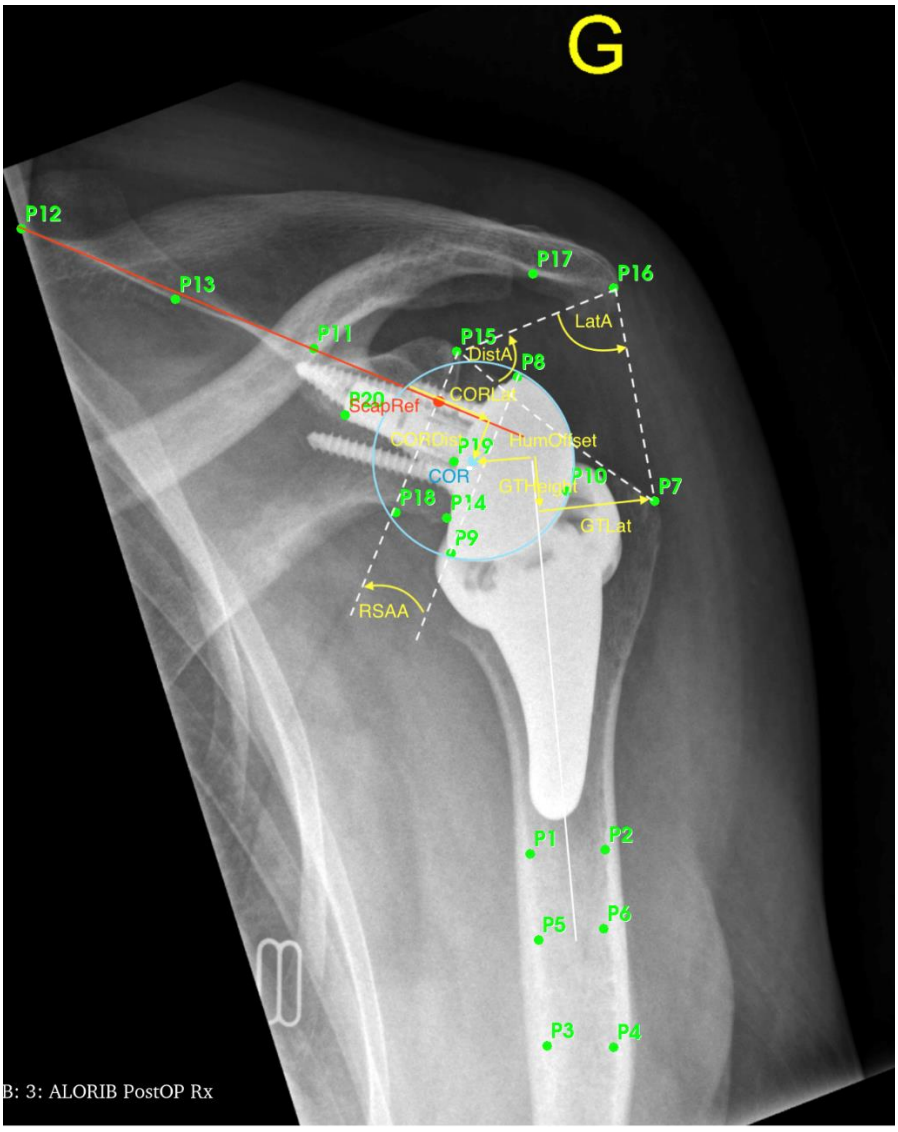
P14, P15 — Lower glenoid / Upper glenoid
Place P14 at the most inferior and P15 at the most superior aspect of the glenoid rim. They define the glenoid line.

P16, P17 — Lateral acromion / Medial acromion (inferior border)
Place two points at the lateral (P16) and medial (P17) edges of the inferior acromial border. They define the acromial line.

P18 — Scapular neck (guideline junction)
Once P17 is placed, a guide line appears: it passes through P14 and is parallel to the fossa axis. Place P18 where this line crosses the cortical edge of the scapular neck.

Reverse prosthesis (RSA): what differs

- **P8, P9, P10 — Glenosphere: P8 upper limit, P9 lower limit, and P10 on the glenosphere surface, widely spaced along its contour. They define the circle of the glenosphere, whose center is the COR.**
- **P14 — Lower reamed glenoid (P15 unchanged).**
- **P18 — Scapular neck: the guide line is the *perpendicular* to the fossa axis passing through P8, not a parallel through P14. Place P18 where it crosses the cortical edge of the scapular neck.**
- **P19, P20 — Peg start / end: place both along the inferior edge of the component. P19 is the limit between the glenoid sphere and the peg, and P20 is the end of the peg.**



For the reverse prosthesis the landmarks should be placed as shown above. Points P1 to P7 do not change.